Sticky OS — Custom reTerminal Sticky Firmware

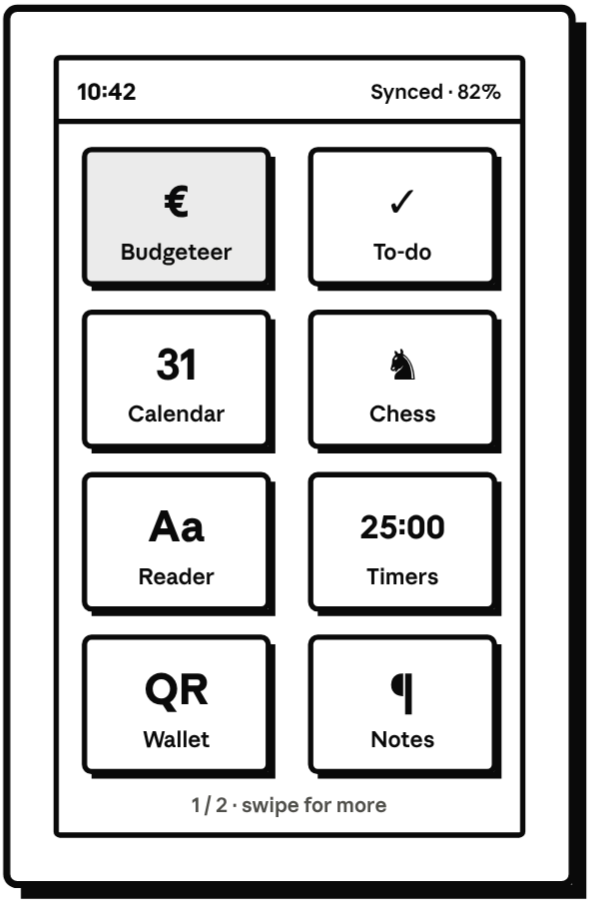
Oct 26, 2026 · @Francesco

What it is

Sticky OS is a custom firmware that turns the Seeed reTerminal Sticky into a pocket device for everyday things: money, tasks, reading, learning, chess, small utilities and a remote terminal. It is built on a fork of the Crosspoint e-reader firmware.

Sticky OS puts every daily tool on one pocket e-paper screen

The launcher in portrait, as drawn on the 480×800 four-gray display



Sticky OS launcher · page 1 of 2

- **Offline-first:** every app except Terminal works with no network. Wi-Fi (home or phone hotspot) is only used to sync.
- **Glanceable:** e-paper, high contrast, portrait, one main action per screen.

- **Configured on the device:** Settings decides which apps appear, their order and icon size.
- **Unlocked:** no PIN or password; whoever holds the device can use it.
- **Not an assistant:** no voice, no AI on the device, no notifications fighting for attention.

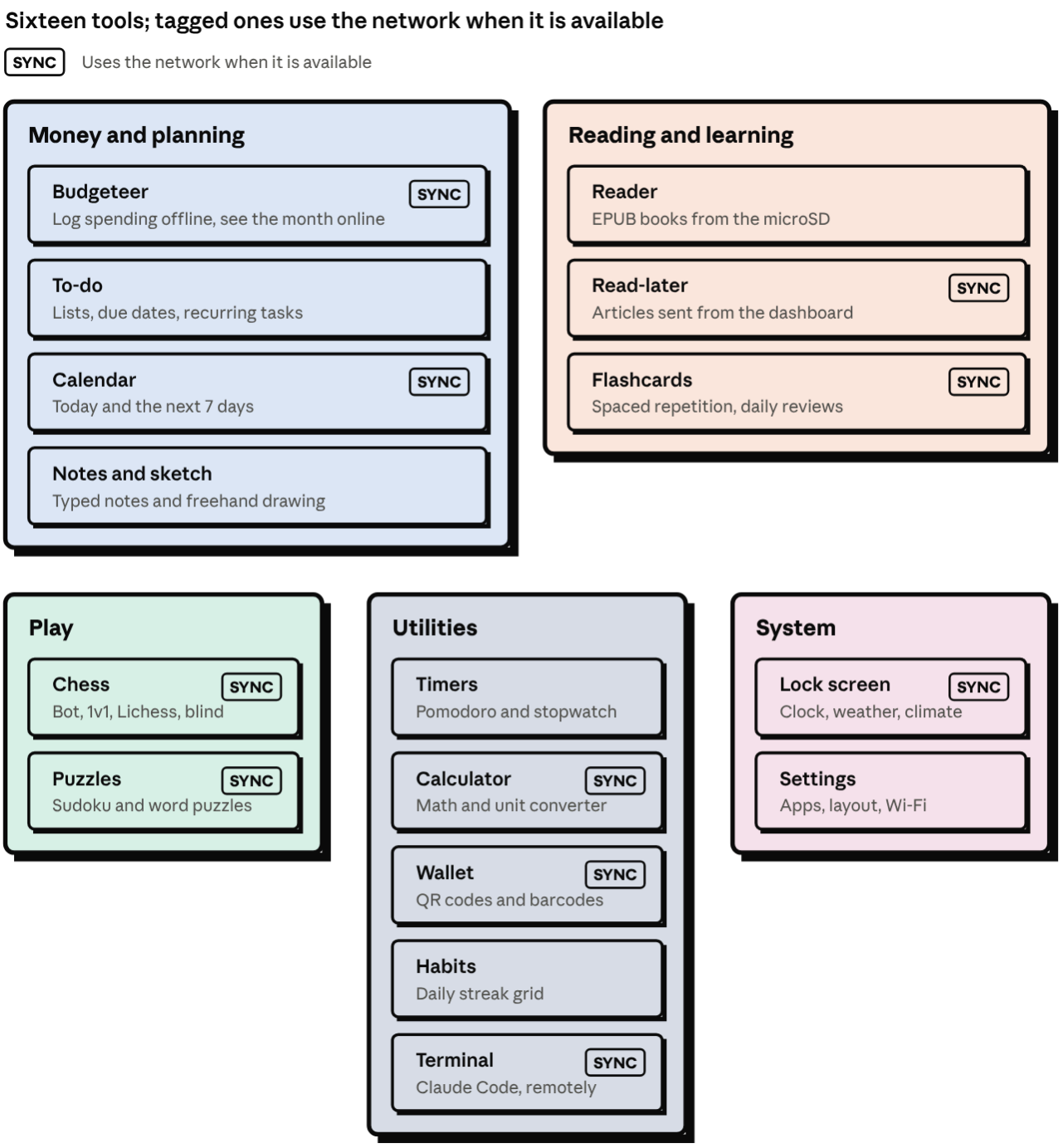
Design language

Sticky OS is neo-brutalist, the style that looks best on a 4-level grayscale e-paper screen. Hard edges and high contrast stay crisp through partial refreshes, and nothing depends on animation. Every screen is designed for portrait, 480×800.

- **Thick outlines:** every card, tile and button has a 3 px black border.
- **Hard shadows:** solid black, offset 4 px down-right, no blur. A pressed button loses its shadow and shifts 4 px, so taps feel physical without animation.
- **Flat fills:** only the four gray levels (white, light gray, dark gray, black). No gradients, no dithering.
- **Heavy type:** bold grotesque for titles, oversized numerals for money, clocks and timers.
- **Inverted focus:** the one thing that matters on a screen (balance, running timer, whose move) is white on black.
- **Big targets:** at least 64×64 px, and each screen's main action is one full-width block.
- **Status bar:** time, battery, Wi-Fi and sync state on every screen.

The app lineup

Sticky OS ships sixteen tools in five groups. All but Terminal work offline; the tagged ones use the network when it is available.



App lineup · 16 tools in 5 groups

Screens

Every screen does one job and ends in one full-width action, in portrait at 480×800.

Main apps in portrait: one job per screen, one big action

Sample data, drawn in the four grays of the 480×800 display



App screens · portrait, sample data

Apps in detail

Every app works offline except Terminal. Apps that use synced data fetch it when a network is available and keep the last copy on the microSD.

Money and planning

Budgeteer. One app with two states.

- **Offline:** only the **+ New transaction** button and a counter of queued entries ("3 to sync").
- **Online:** opening the app sends the queue, then loads the month. Home shows income, expenses and balance; tabs show spent vs. budget per category and the transaction list, with arrows to change month.

Adding a transaction takes six steps:

1. **Type:** Expense or Income.
2. **Amount:** numeric keypad with decimal comma.
3. **Category:** grid filtered by type.
4. **Description:** chips with the last 5 descriptions for that category, the keyboard, or Skip.
5. **Date:** Today by default, with Yesterday and ±1 day.
6. **Save:** sent at once if online, otherwise queued.

Categories arrive with the synced data, so the device never hardcodes them. Income categories are Stipendio and Introito2; expense categories are Equitalia, Bollette, Trasporti, Spesa, Svago, Ristoranti, Risparmio, Abbonamenti, Salute, Cura and NoCat, with NoCat kept out of the monthly balance. A queued entry leaves the queue only once the backend confirms it, so a lost confirmation can cause a rare duplicate.

To-do. Lists, due dates, recurring tasks and tap-to-complete, typed on a compact on-screen keyboard. Data stays on the device.

Calendar. A read-only agenda for today and the next 7 days, from synced data. Offline, it shows the last copy.

Notes and sketch. Typed notes and freehand touch drawing on partial refresh. Stored on the device.

Reading and learning

Reader. The Crosspoint e-reader: EPUB from microSD, bookmarks, saved position, adjustable font and margins, page turns by button or swipe.

Read-later. Articles arrive on sync as clean text and open in the reader engine. Marking one as read is sent back on the next sync.

Flashcards. Decks arrive on sync or are copied to the microSD as CSV. Spaced-repetition scheduling runs on the device, with a daily review count.

Play

Chess. Four modes:

- **Vs bot:** an offline lightweight engine with adjustable strength.
- **1v1:** pass-and-play, the board flips after every move, optional clock.
- **Lichess:** correspondence, classical and rapid games plus the daily puzzle, using your personal API token typed once in Settings. No blitz, because of e-paper refresh times.
- **Blindfold trainer:** square-color and coordinate drills, a 15-minute daily session and a streak counter.

Puzzles. Sudoku generated offline in several difficulties, plus word puzzles: a bundled pack and a daily puzzle that arrives on sync.

Utilities

Timers. Pomodoro (configurable work and break lengths), stopwatch with laps, and countdown. Timers keep running through sleep; the end is shown on screen.

Calculator. Standard calculator plus unit converter (length, weight, volume, temperature, currency). Currency rates arrive on sync and are cached.

Wallet. Loyalty cards, tickets and boarding passes. Codes arrive on sync as text with a label, and the device draws each QR code or barcode itself at full contrast.

Habits. A daily checklist with a streak grid of the last weeks. Data stays on the device.

Terminal. A terminal-style client for Claude Code sessions running on your computer. It needs a network and works live rather than through the sync queue.

- Monospace transcript of the session, scrolled with the buttons.
- Quick-reply chips (continue, run the tests, explain) and the on-screen keyboard for short prompts.
- Permission requests shown as full-width Approve and Deny buttons.

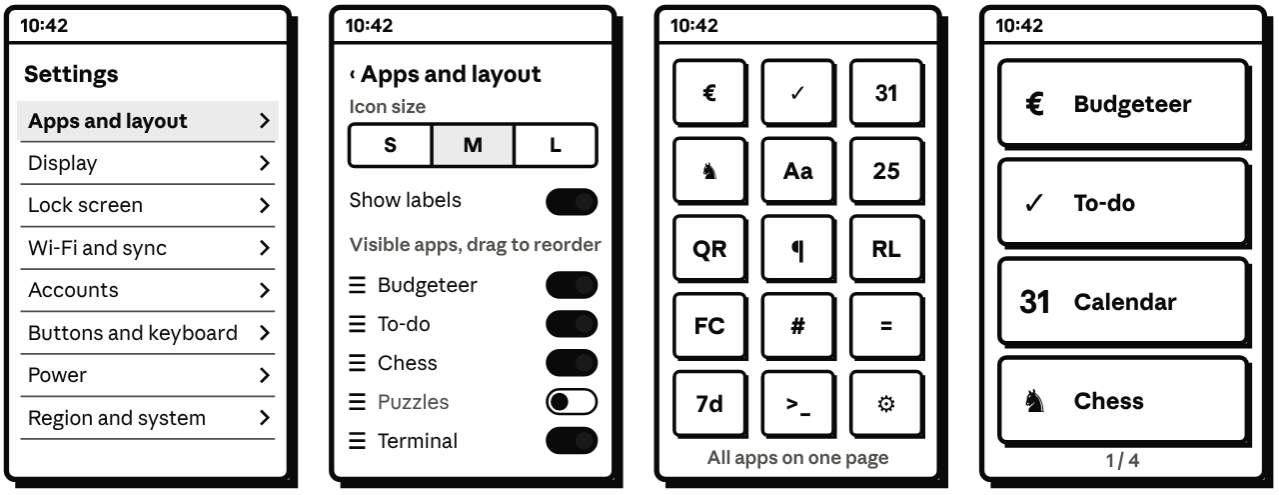
System

Lock screen. The idle screen: clock, date, synced weather, and room temperature and humidity from the onboard sensor. It is not a lock; there is no PIN or password. Its refresh interval and items are set in Settings.

Settings. A full app for everything the device does. It is always in the launcher and can't be hidden.

Settings controls which apps show and how big their icons are

Settings, Apps and layout, and the launcher at small and large icon size



Settings · main menu, apps and layout, icon sizes

- **Apps and layout:** show or hide each app, reorder them, pick the icon size (small 3×5 grid, medium 2×4, large single column), and turn labels on or off.
- **Display:** system font size, full-refresh interval (cleaner image vs. faster screens), inverted colors.
- **Lock screen:** refresh interval, clock style, and which items show (weather, room climate, next event).
- **Wi-Fi and sync:** saved networks with priority, auto-sync on or off, sync now, last sync time.
- **Accounts:** Lichess API token, backend address and key.
- **Buttons and keyboard:** button mapping, keyboard layout, accented letters.
- **Power:** sleep timeout, battery level, battery saver.
- **Region and system:** 24-hour clock, date format, decimal comma, metric units, storage per app, clear caches, firmware version and update check, factory reset.

Hardware

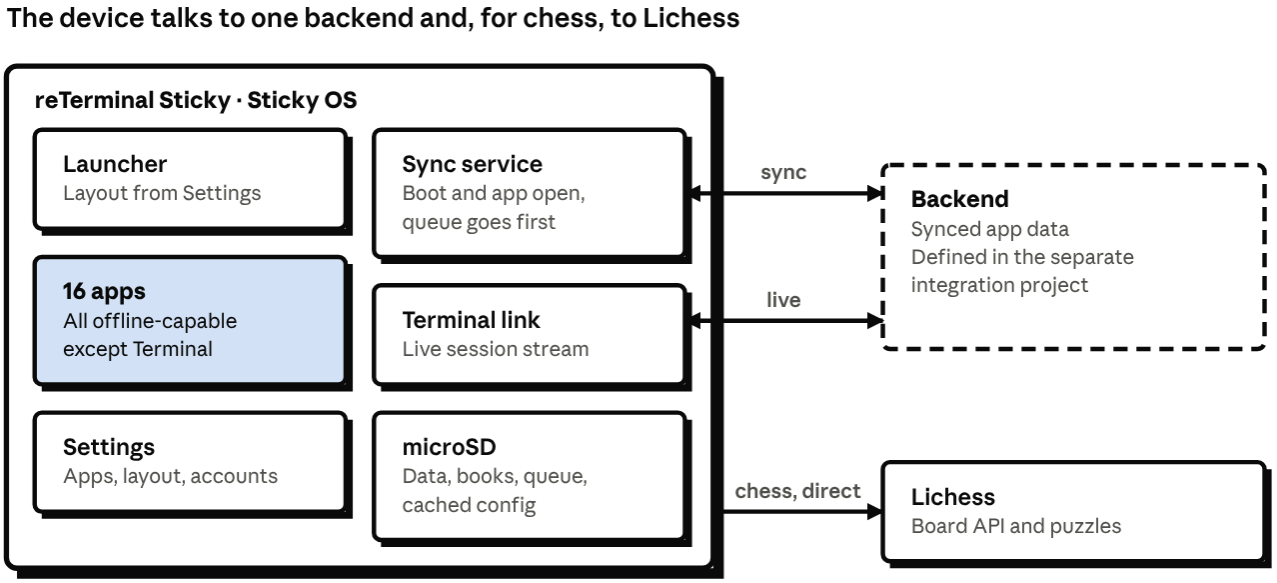
Sticky OS runs on the stock Seeed reTerminal Sticky, with no hardware changes.

Component	Spec	How Sticky OS uses it
SoC	ESP32-S3R8, 8 MB PSRAM	Lightweight chess engine instead of Stockfish

Component	Spec	How Sticky OS uses it
Display	3.97-inch e-paper, 800×480, 4-level grayscale, capacitive touch	Partial refresh in apps, full refresh every N updates
Storage	32 MB flash + microSD	All data, books, articles, decks and caches on the microSD
Connectivity	2.4 GHz Wi-Fi, BLE 5.0	Sync over home Wi-Fi or phone hotspot; BLE unused
Battery	750 mAh, USB-C	Deep sleep between interactions
Sensors	Temperature/humidity, 3-axis accelerometer	Room climate on the lock screen; wake on tap
Audio	Single PDM microphone, no speaker	Unused
Input	Touch + physical buttons	Buttons for home, back and page turns

How it works

All synced data moves over a single backend connection, defined in a separate integration project. Lichess is the only service the device reaches directly.



How it works · device, backend, Lichess

- **Launcher:** an app grid with only the enabled apps, in the order and icon size set in Settings. Physical buttons handle home, back and page turns.

- **Storage:** one folder per app on the microSD, plus the Budgeteer queue, synced content and the settings.
- **Sync:** at boot and when an app that needs data opens, over home Wi-Fi or a phone hotspot. Queued entries go first, then fresh data comes down.
- **Power:** deep sleep between interactions, partial refresh inside apps, a full refresh every N updates.
- **Updates:** over the air from GitHub releases.

Sync and configuration

The device expects one backend that hands it per-app data, and accepts a few things back. How that backend is built is out of scope here.

Area	Device receives	Device sends
Budgeteer	Categories, month transactions, totals and budgets	New transactions
Calendar	Agenda for 7 days	—
Read-later	Articles as clean text	Read status
Flashcards	Decks	—
Puzzles	Daily word puzzle	—
Calculator	Currency rates	—
Wallet	Code payloads and labels	—
Lock screen	Weather	—
Terminal	Session output and permission requests, live	Prompts and approvals, live

Rules:

- Settings live only on the device; layout and app visibility are never synced.
- Hidden apps stay in the firmware and keep their data, so showing one again is instant.
- Each app's last synced copy stays on the microSD for offline use.

To verify during the build

- [ ] Number of physical buttons and their mapping.

- [ ] Whether the Sticky has any buzzer or vibration for timer alerts.
- [ ] Crosspoint's license, and whether it can host a multi-app launcher.
- [ ] Memory headroom on 8 MB of PSRAM with the chess engine, reader, Wi-Fi stack and Terminal's live connection.
- [ ] Battery life with lock-screen refreshes and syncs, to tune the defaults.
- [ ] Chess engine choice and its strength levels.
- [ ] Terminal readability: how many columns fit at a readable font size.